

## Tutorial 12

### Diagonalization and applications

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1. For each of the following statements, say whether it is true or false. Give a reason for your answer,

- (a) Every diagonalizable matrix is invertible.

**Solution:** False.

For instance the matrix  $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$  is diagonalizable but not invertible.

- (b) Every invertible matrix is diagonalizable.

**Solution:** False.

Take for instance the matrix

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

We can see that  $A$  is invertible. But  $A$  is not diagonalizable because  $A$  does not have two linearly independent eigenvectors.

- (c) If a square matrix has a repeated eigenvalue, it is not diagonalizable.

**Solution:** False.

For instance the identity matrix  $I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$  has the repeated eigenvalue  $\lambda = 1$ , but  $I_2$  is diagonalizable.

2. In Tutorial 11, you found eigenvalues and bases for the corresponding eigenspaces for three matrices, in Questions 3, 5 and 6. Use your answers to those questions to determine whether the matrices are diagonalizable; give a reason in each case. In the case(s) where the matrix is diagonalizable, give an invertible matrix  $P$  which diagonalizes the matrix, and the corresponding diagonal matrix  $D$ .

- (a) The matrix  $A$  in Question 3.

**Solution:**  $A$  is diagonalizable since it has 3 linearly independent eigenvectors.

$$P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix} \quad D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

- (b) The matrix  $A$  in Question 5.

**Solution:**  $A$  is not diagonalizable, since it has only 2 linearly independent eigenvectors.

- (c) The matrix  $B$  in Question 6.

**Solution:**  $B$  is diagonalizable, since  $\mathbb{R}^3$  has a basis consisting of eigenvectors of  $B$ .

$$P = \begin{pmatrix} 4 & 1 & -1 \\ 1 & 0 & 4 \\ -1 & 4 & 0 \end{pmatrix}, \quad D = \begin{pmatrix} 9 & 0 & 0 \\ 0 & -9 & 0 \\ 0 & 0 & -9 \end{pmatrix}.$$

3. Let  $A = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 3 & 1 \\ 0 & -3 & -1 \end{pmatrix}$ .

- (a) Find the eigenvalues of  $A$ .

**Solution:** The eigenvalues are 0, 1 and 2.

- (b) Use only your answer to (a) to prove or disprove the following statements:
- $A$  is diagonalisable.
  - $A$  is invertible.

**Solution:**

- True. The  $3 \times 3$  matrix  $A$  has 3 distinct eigenvalues. The corresponding eigenvectors are linearly independent.
- False. Since 0 is an eigenvalue,  $A$  is not invertible.

- (c) If  $A$  is diagonalisable, find an invertible matrix  $P$  and a diagonal matrix  $D$  such that  $P^{-1}AP = D$ .

**Solution:**  $P = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ -3 & 0 & -1 \end{pmatrix}, \quad D = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$

(This is not the only correct answer.)

- (d) Prove or disprove:
- There is a  $k \in \mathbb{N}$  such that  $A^k$  is the zero matrix.
  - There is a  $k \in \mathbb{N}$  such that  $A^k = I$ .

**Solution:**

- i. False. From  $P^{-1}AP = D$  we get  $A = PDP^{-1}$  and hence for any  $k \in \mathbb{N}$ ,  $A^k = PD^kP^{-1}$ .

If there were a  $k \in \mathbb{N}$  such that  $A^k = \mathbf{0}$ , we would have  $PD^kP^{-1} = \mathbf{0}$ , and so  $D^k = P^{-1}\mathbf{0}P = \mathbf{0}$ , a contradiction since  $D^k \neq \mathbf{0}$  for every  $k$ .

- ii. False. From  $P^{-1}AP = D$  we get  $A = PDP^{-1}$  and hence for any  $k \in \mathbb{N}$ ,  $A^k = PD^kP^{-1}$ .

If there were a  $k \in \mathbb{N}$  such that  $A^k = I$ , we would have  $PD^kP^{-1} = I$ , and so  $D^k = P^{-1}IP = I$ , a contradiction since  $D^k \neq I$  for every  $k$ .

4. In this question we use diagonalization to solve a system of linear differential equations. You will need to know that the general solution of the differential equation  $x'(t) = \frac{dx}{dt} = kx$  is  $x(t) = Ce^{kt}$ , with  $C$  a constant.

Consider the system of differential equations:

$$\begin{aligned}x_1'(t) &= x_1(t) + 3x_2(t) \\x_2'(t) &= 4x_1(t) + 5x_2(t).\end{aligned}$$

- (a) Write  $\mathbf{x}(t)$  for  $\begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix}$  and  $\mathbf{x}'(t)$  for  $\begin{pmatrix} x_1'(t) \\ x_2'(t) \end{pmatrix}$ .

Use this notation to write system above in the form

$$\mathbf{x}'(t) = A\mathbf{x}(t), \quad (*)$$

for some matrix  $A$ .

**Solution:** The system can then be written in the form

$$\mathbf{x}'(t) = \begin{pmatrix} 1 & 3 \\ 4 & 5 \end{pmatrix} \mathbf{x}(t). \quad (*)$$

- (b) Find an invertible matrix  $P$  and a diagonal matrix  $D$  such that  $P^{-1}AP = D$ .

**Solution:** The matrix  $A = \begin{pmatrix} 1 & 3 \\ 4 & 5 \end{pmatrix}$  has eigenvalues 7 and  $-1$  with corresponding eigenvectors  $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$  and  $\begin{pmatrix} -3 \\ 2 \end{pmatrix}$ .

If we put  $P = \begin{pmatrix} 1 & -3 \\ 2 & 2 \end{pmatrix}$  and  $D = \begin{pmatrix} 7 & 0 \\ 0 & -1 \end{pmatrix}$ , then  $P^{-1}AP = D$ .

- (c) Let  $\mathbf{x}(t) = P\mathbf{u}(t)$ , where  $\mathbf{u}(t) = \begin{pmatrix} u_1(t) \\ u_2(t) \end{pmatrix}$ . Show that  $\mathbf{x}'(t) = P\mathbf{u}'(t)$ , where  $\mathbf{u}'(t) = \begin{pmatrix} u_1'(t) \\ u_2'(t) \end{pmatrix}$ . Substitute in equation (\*).

**Solution:** Write the vector equation  $\mathbf{x}(t) = P\mathbf{u}(t)$  as two linear equations, differentiate each of these equations and write the result as a vector equation again. Then equation (\*) becomes

$$P\mathbf{u}'(t) = AP\mathbf{x}(t) \text{ or } \mathbf{u}'(t) = P^{-1}AP\mathbf{x}(t) = D\mathbf{u}(t).$$

- (d) You should now have two simple differential equations for  $u_1$  and  $u_2$ . Solve them.

**Solution:** In terms of components this becomes

$$\begin{aligned} u_1'(t) &= 7u_1(t) \\ u_2'(t) &= -u_2(t). \end{aligned}$$

These two equations give the solutions  $u_1(t) = C_1e^{7t}$ ,  $u_2(t) = C_2e^{-t}$ , where  $C_1$  and  $C_2$  are arbitrary constants.

- (e) Use your solutions for  $u_1$  and  $u_2$  to find solutions for  $x_1$  and  $x_2$ .

**Solution:** We have  $\mathbf{x}(t) = P\mathbf{u}(t)$ , or

$$\begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} = \begin{pmatrix} 1 & -3 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} u_1(t) \\ u_2(t) \end{pmatrix} = \begin{pmatrix} 1 & -3 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} C_1e^{7t} \\ C_2e^{-t} \end{pmatrix},$$

so

$$x_1(t) = C_1e^{7t} - 3C_2e^{-t} \text{ and } x_2(t) = 2C_1e^{7t} + 2C_2e^{-t}.$$

5. Excessive university fees have driven you to selling subscriptions to a pathetic magazine entitled “How to Succeed in Linear Algebra” to make ends meet. Every month, you send an email to a large mailing list. Some of the students receiving this email already subscribe to your magazine, others do not. You notice the following: from the mailing list, 60 percent of those who already subscribe will subscribe again, while 25 percent of those who do not now subscribe will subscribe.

- (a) Write down the transition matrix  $M$  for this process: that means the matrix  $M$  satisfying  $\mathbf{u} = M\mathbf{x}$  where

$$\begin{aligned} \mathbf{u} &= \begin{pmatrix} \text{fraction of people on mailing list subscribing after letter} \\ \text{fraction of people on mailing list not subscribing after letter} \end{pmatrix} \\ \mathbf{x} &= \begin{pmatrix} \text{fraction of people on mailing list subscribing before letter} \\ \text{fraction of people on mailing list not subscribing before letter} \end{pmatrix} \end{aligned}$$

**Solution:**  $M = \begin{pmatrix} 0.6 & 0.25 \\ 0.4 & 0.75 \end{pmatrix}$

- (b)  $M$  should be a Markov matrix: is it?

**Solution:** Yes.

- (c) Find an invertible matrix  $P$  and a diagonal matrix  $D$  such that  $P^{-1}MP = D$ . [Shortcut: a Markov matrix always has 1 as eigenvalue; what is the other eigenvalue?].

**Solution:** Do the usual eigenvalue and eigenvector calculations. You get eigenvalues 1 and 0.35. The corresponding eigenspaces are

$$W_1 = \left\{ \begin{pmatrix} 5 \\ 8 \end{pmatrix} t : t \in \mathbb{R} \right\} \text{ and } W_{0.35} = \left\{ \begin{pmatrix} 1 \\ -1 \end{pmatrix} t : t \in \mathbb{R} \right\}.$$

One can use  $P = \begin{pmatrix} 5 & 1 \\ 8 & -1 \end{pmatrix}$  and  $D = \begin{pmatrix} 1 & 0 \\ 0 & 0.35 \end{pmatrix}$ .

- (d) Suppose you begin with nobody subscribing. According to this model, what fraction of the people on the mailing list will subscribe after your first letter?

**Solution:**  $M \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0.25 \\ 0.75 \end{pmatrix}$ , so one quarter of the people on the list will subscribe after one letter.

- (e) Use  $M = PDP^{-1}$  to find an expression for  $M^n$  in terms of  $P, P^{-1}$  and  $D^n$ . Find  $A = \lim_{n \rightarrow \infty} M^n$ . What does this mean for your subscription situation in the long run?

**Solution:**  $A = \begin{pmatrix} \frac{5}{13} & \frac{5}{13} \\ \frac{8}{13} & \frac{8}{13} \end{pmatrix}$ . In the long run,  $\frac{5}{13}$  of your list will subscribe.

- (f) The columns of  $A$  should all be the same. Let  $\mathbf{v}$  be one of these columns. What is  $M\mathbf{v}$ ?

**Solution:**  $M\mathbf{v} = \mathbf{v}$  because  $\mathbf{v}$  is an eigenvector of  $M$  corresponding to the eigenvalue 1.

6. Let  $A$  be an  $n \times n$  real matrix. We call a subspace  $S$  of  $\mathbb{R}^n$  **invariant** under the matrix  $A$  if  $A\mathbf{x} \in S$  for every  $\mathbf{x} \in S$  (i.e.  $A$  sends every vector in  $S$  to a (possibly different) vector in  $S$ , never to a vector outside  $S$ ). The trivial subspace  $\{\mathbf{0}\}$  and  $\mathbb{R}^n$  itself are always invariant subspaces. Show a non-trivial proper subspace of  $\mathbb{R}^2$  is an invariant subspace of a  $2 \times 2$  matrix  $A$  if and only if it is an eigenspace of  $A$ .

**Solution:** An eigenspace of a  $2 \times 2$  matrix  $A$  which is also a proper subspace of  $\mathbb{R}^2$  must be 1-dimensional and therefore of the form  $S = \{\mu \mathbf{u} : \mu \in \mathbb{R}\}$ , with  $\mathbf{u}$  an eigenvector of  $A$ . Let  $\lambda$  be the corresponding eigenvalue. If  $\mathbf{v} \in S$ , then  $\mathbf{v} = \mu \mathbf{u}$  and  $A\mathbf{v} = A(\mu \mathbf{u}) = \mu A\mathbf{u} = \mu \lambda \mathbf{u} \in S$ . Hence  $S$  is an invariant subspace of  $A$ . Conversely, a subspace of  $\mathbb{R}^2$  other than  $\{\mathbf{0}\}$  or  $\mathbb{R}^2$  must be a line through the origin, i.e. it must be a subspace of the form  $\{\mu \mathbf{u} : \mu \in \mathbb{R}\}$ , for some non-zero  $\mathbf{u} \in \mathbb{R}^2$ . If this subspace is invariant for  $A$ , then we must have  $A\mathbf{u} = \mu \mathbf{u}$  for some  $\mu \in \mathbb{R}$ . But then  $\mathbf{u}$  must be an eigenvector and  $\mu$  an eigenvalue for  $A$ . From this it follows that the subspace is an eigenspace of  $A$ .